

**An Algebraic Invariance Condition for the Existence of Latent Correlations in
Confirmatory Factor Analysis**

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Version 2, Sep 2026.

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Abstract

An independent cluster model in confirmatory factor analysis (ICM-CFA) assigns each variable to one and only one factor. Variables loading on the same factor form a cluster representing a construct, and correlations among factors represent latent correlations among these constructs. We derive the covariance condition required for a population latent correlation to exist between two clusters. The within- and between-cluster covariance patterns must align up to a common scalar. Each admissible quadruple identifies an alignment ratio, and under ICM-CFA all alignment ratios equal the squared latent correlation. When they differ, no single population latent correlation exists, and fitting ICM-CFA returns a criterion-dependent summary rather than an estimate of a common population quantity. With more than two clusters, this condition applies pairwise. We investigate two forms of misspecification: misassignment, when variables are assigned to the wrong cluster, and misalignment, when correctly assigned covariance patterns do not align. Alignment ratios diagnose both forms of misspecification without additional model fitting, and their invariance condition clarifies the relationship between ICM-CFA and bifactor models.

Keywords: Confirmatory Factor Analysis, Structural Equation Modeling, Latent Factor Correlations, Model Misspecification, Bifactor Model, Alignment

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Introduction

Confirmatory factor analysis (CFA) is a standard framework in psychological research for relating task performance to putative constructs. One of the most popular implementations has the following constraint. Each task is *a priori* assigned to one and only one of several factors. Tasks that load on the same factor form a cluster, and no task may be assigned to two clusters. Models with this cluster property appear at least as early as Thurstone (1947) and they go by several names. We follow Marsh et al. (2014) who call it the *independent clusters model of confirmatory factor analysis*, and we refer to it herein by the acronym ICM-CFA. An example of an ICM-CFA, from Tsukahara et al. (2020), is shown in Figure 1A where tasks are clustered into cognitive factors of *fluid intelligence*, *working memory*, *attentional control*, and *sensory discrimination*. ICM-CFA is an oblique model that allows for correlations between factors. For example, the correlation between attentional control and sensory discrimination is 0.9 (see red box), and this correlation is interpreted by Tsukahara et al. (2020) as indicating that limitations in attentional control result in poorer sensory discrimination.

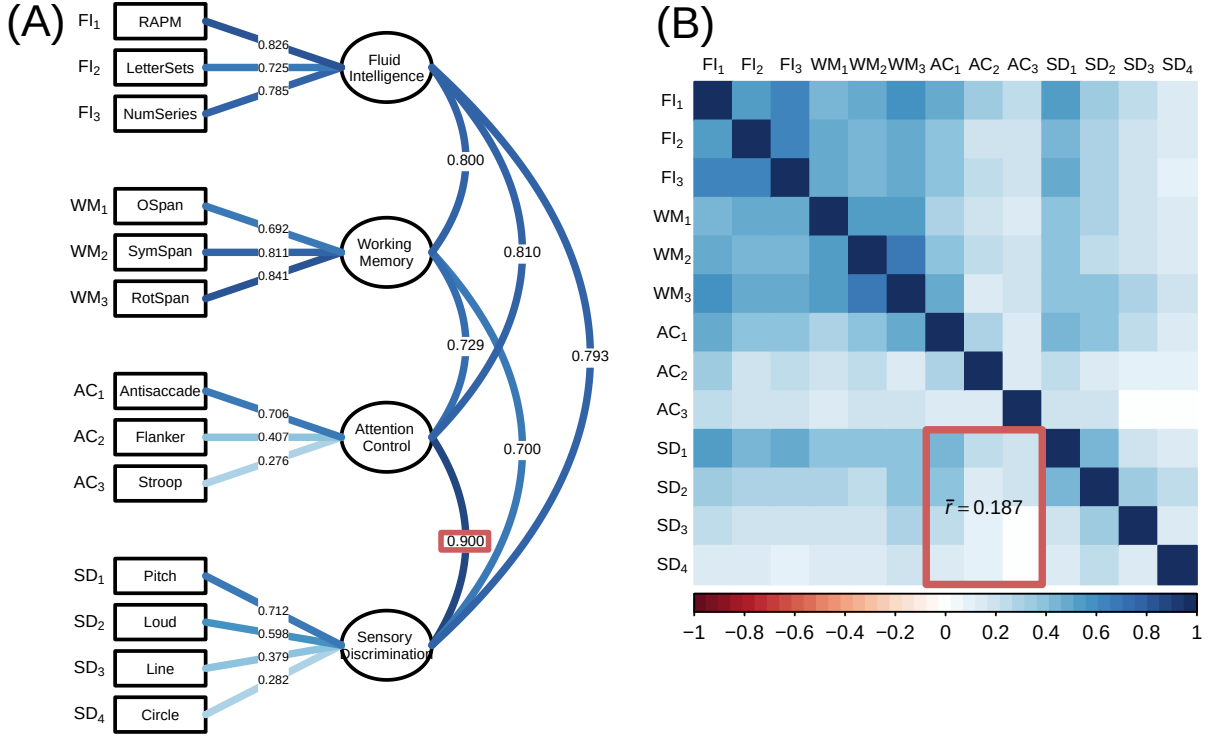
The key in most implementations is that the grouping of tasks into clusters is made *a priori* on theoretical grounds. In Figure 1, for example, the authors decided *a priori* to load the Stroop task, the flanker task, and the antisaccade task on the attentional factor and on no other factor. The ICM-CFA model is sometimes criticized as overly restrictive because it fixes all cross-loadings to zero (Asparouhov & Muthén, 2009; Marsh et al., 2014; Morin et al., 2016). When cross-loadings are nonzero but treated as zero, latent-correlation estimates may be inflated (Asparouhov et al., 2015; Marsh et al., 2009; Steenkamp & Maydeu-Olivares, 2023). The reason is that the shared variance across variables in different clusters must be represented, and the model may only do so through the latent correlations. This inflation

has been documented in the exploratory structural equation modeling (ESEM) literature in applications and in Monte Carlo studies (Hsu et al., 2014; Marsh et al., 2013, 2020; Tóth-Király et al., 2017; van Zyl & ten Klooster, 2022). Here, we provide a more formal analysis.

The concern about inflated correlations applies to the example in Figure 1. The manifest correlations across the tasks are shown in Figure 1B. These are small in value. The correlations between sensory discrimination and attentional control tasks, shown in the red box, average only 0.19. This value is surprisingly small given that the fitted latent correlation between the sensory discrimination and attentional control constructs is 0.9. It is conceivable that this large discrepancy reflects inflation due to misspecification.

In this paper, we present a novel analysis of ICM-CFA for interpreting latent correlations. We show that ICM-CFA imposes algebraic constraints on the manifest covariance matrix among tasks. In a properly specified ICM-CFA model, diagonal blocks of the covariance matrix describe within-cluster relations, whereas off-diagonal blocks describe between-cluster relations. The ICM-CFA model implies specific relations between the within-cluster and between-cluster submatrices that we term *alignment*. In general, each block may have its own pattern, but if ICM-CFA holds, products of between-cluster covariance entries must match appropriate products of within-cluster covariances up to a scalar. The scalar implied by each admissible quadruple describes the alignment, and we call it the *alignment ratio*. In ICM-CFA, these alignment ratios are invariant across admissible task quadruples, and the common value is the squared latent correlation.

Alignment ratios provide a novel understanding of ICM-CFA and its close relationship to the bifactor model. We use this understanding to analyze the effects of misspecification. There are two qualitatively different types of misspecification. One has been studied and occurs when the analyst misassigns a task. In truth, the task has a nonzero loading on a different factor than the one to which it is assigned. With the benefit of

**Figure 1**

ICM-CFA solution and manifest task correlations from Tsukahara et al. (2020). (A) Fitted four-factor ICM-CFA solution with standardized task loadings and latent factor correlations, using the authors' original specification. The highlighted example shows a large fitted latent correlation between attentional control and sensory discrimination. (B) Manifest task correlation matrix. The highlighted between-cluster region corresponds to pairs of attentional control and sensory discrimination tasks, with mean correlation 0.19, illustrating the gap between low task correlations and high fitted latent correlations.

alignment ratios, we show that, with two clusters, misassignment cannot move an alignment ratio below its correctly specified value, so the distortion is one-sided. With three or more clusters, however, the direction of the distortion depends on the broader pattern of latent correlations and misassignment.

There is also a second type of misspecification. The analyst may assign tasks to clusters correctly, but the within-cluster and between-cluster submatrices may still fail to

align. In this case, alignment ratios are not invariant, and no single population latent correlation captures the relations across clusters. ICM-CFA nonetheless returns a fitted latent correlation. We use the square root of the geometric mean of the population alignment ratios as a reference for understanding this fitted value. We show that, across the numerical study, greater misalignment tends to produce a larger upward displacement of the fitted latent correlation relative to this reference.

ICM-CFA Model Specification

Let $\mathbf{y}_i = (y_{i1}, \dots, y_{iJ})'$ be a column vector of J observations for the i th individual. In cognitive psychology, for example, these observations are performance scores in a battery of tasks. We model these observations as a multivariate normal:

$$\mathbf{y}_i \sim \mathcal{N}_J(\boldsymbol{\mu}, \boldsymbol{\Sigma}),$$

where $\boldsymbol{\mu} = (\mu_1, \dots, \mu_J)'$ is a column vector of means and $\boldsymbol{\Sigma}$ is a $J \times J$ covariance matrix. As usual, the covariance may be expressed as a function of correlation matrix $\boldsymbol{\rho}$:

$$\boldsymbol{\Sigma} = \text{diag}(\boldsymbol{\sigma}) \boldsymbol{\rho} \text{diag}(\boldsymbol{\sigma}),$$

where $\text{diag}(\boldsymbol{\sigma})$ is a diagonal matrix with diagonal elements $\sigma_1, \dots, \sigma_J$.

This covariance matrix is the object of interest, and in SEM it is decomposed as

$$\boldsymbol{\Sigma} = \boldsymbol{\Lambda}\boldsymbol{\Phi}\boldsymbol{\Lambda}' + \boldsymbol{\Psi}.$$

Here $\boldsymbol{\Lambda}$ is the $J \times K$ loading matrix relating J observed tasks to K latent factors. The $K \times K$ factor covariance matrix $\boldsymbol{\Phi}$ encodes covariation among the factors, and $\boldsymbol{\Psi} = \text{diag}(\psi_1, \dots, \psi_J)$ is the $J \times J$ diagonal task-specific residual covariance matrix. The implication of this diagonal formulation is that residuals are mutually uncorrelated across tasks. This decomposition separates $\boldsymbol{\Sigma}$ into a common component, $\boldsymbol{\Lambda}\boldsymbol{\Phi}\boldsymbol{\Lambda}'$, that captures shared covariation among tasks and a task-specific component, $\boldsymbol{\Psi}$, that captures the remaining task variance not explained by the common part. Factor models are then specified

by placing restrictions on $\mathbf{\Lambda}$ (and sometimes on $\mathbf{\Phi}$), yielding either confirmatory (restricted) or exploratory (less restricted) formulations.

ICM-CFA is a restricted factor model in which each task loads on one and only one factor. For example, with $J = 6$ tasks and $K = 2$ factors, if Tasks 1 through 3 are taken to measure one construct and Tasks 4 through 6 another, the loading matrix may be specified as:

$$\mathbf{\Lambda} = \begin{bmatrix} \lambda_1 & 0 \\ \lambda_2 & 0 \\ \lambda_3 & 0 \\ 0 & \lambda_4 \\ 0 & \lambda_5 \\ 0 & \lambda_6 \end{bmatrix}.$$

This specification of $\mathbf{\Lambda}$ encodes a perfect-cluster structure where tasks load on their assigned factor and off-factor loadings are fixed to zero. Covariation between tasks in different clusters is therefore attributed to association between the two latent factors, encoded in the factor covariance matrix $\mathbf{\Phi}$.

There are two equivalent parameterizations commonly used to set the scale of the model. We describe them using the six-task, two-factor example above. Both apply to any number of tasks and factors by imposing the same identifying restriction separately for each factor. The first fixes one loading per factor to 1.0; in the present example, $\lambda_1 = \lambda_4 = 1$. This sets the factor scales while leaving the factor variances free to be estimated. In the second parameterization each factor variance is fixed to 1.0. No loadings are fixed, and the factor covariance matrix is therefore a correlation matrix. For example,

$$\mathbf{\Phi} = \begin{bmatrix} 1 & \varphi \\ \varphi & 1 \end{bmatrix},$$

where φ is the correlation between the two factors. Without loss of generality, we use this second parameterization throughout.

The latent correlation φ may be interpreted as the correlation between latent scores. An equivalent formulation, obtained by introducing latent scores and conditioning on them, is

$$\mathbf{y}_i | \boldsymbol{\eta}_i \sim \mathcal{N}_J(\boldsymbol{\mu} + \boldsymbol{\Lambda} \boldsymbol{\eta}_i, \boldsymbol{\Psi}),$$

where $\boldsymbol{\eta}_i = (\eta_{i1}, \dots, \eta_{iK})'$ is a vector of latent scores for the i th individual, distributed as $\boldsymbol{\eta}_i \sim \mathcal{N}_K(\mathbf{0}, \boldsymbol{\Phi})$. The correlation between scores in the two-factor example is $\text{cor}(\eta_{i1}, \eta_{i2}) = \varphi$.

Latent correlations as corrections for measurement error

In the introduction, we asked why manifest correlations among tasks are low while latent correlations among the factors are high. Here, we provide an interpretation of latent correlations and the role of measurement error. Under classical test theory, Revelle and Condon (2019) note that Spearman’s correction for attenuation (Spearman, 1904) can be interpreted as the correlation between the true scores (the error-free components) of two measures. Here, we derive the same Spearman correction from an explicit latent-variable measurement model and show how it arises under ICM-CFA.

Consider the idealized case in which the tasks (or indicators) within a cluster measure a common latent variable in the same way, apart from additive constants and idiosyncratic measurement error. Such tasks are termed *essentially parallel* (Lord et al., 1968; Widaman & Revelle, 2022). For example, two alternate forms of a working-memory task may differ in difficulty in a way that produces a constant shift in scores, while otherwise measuring working-memory ability with the same strength and having the same error variance.

To formalize this setting, suppose the J task scores $y_{i1}, \dots, y_{iJ}$ for individual i are partitioned into two clusters A and B , with index sets $\mathcal{A}$ and $\mathcal{B}$ of sizes m_A and m_B , where $m_A + m_B = J$. Let η_{iA} and η_{iB} denote the corresponding latent scores, with $\text{cor}(\eta_{iA}, \eta_{iB}) = \varphi$. In a CFA parameterization, essential parallelism restricts tasks within each cluster to a

common loading and a common residual variance, while allowing intercepts to vary:

$$y_{i\ell} = \mu_\ell + \lambda_A \eta_{iA} + \varepsilon_{i\ell}, \quad \ell \in \mathcal{A},$$

$$y_{ik} = \mu_k + \lambda_B \eta_{iB} + \varepsilon_{ik}, \quad k \in \mathcal{B}.$$

Here, μ_j are task intercepts, and ε_{ij} are idiosyncratic measurement errors with $\mathbb{E}(\varepsilon_{ij}) = 0$, $\text{cov}(\varepsilon_{ij}, \varepsilon_{ij'}) = 0$ for $j \neq j'$, and $\varepsilon_{ij} \perp (\eta_{iA}, \eta_{iB})$. Residual variances are constant within each cluster:

$$\text{var}(\varepsilon_{i\ell}) = \psi_A, \quad \ell \in \mathcal{A}, \quad \text{and} \quad \text{var}(\varepsilon_{ik}) = \psi_B, \quad k \in \mathcal{B}.$$

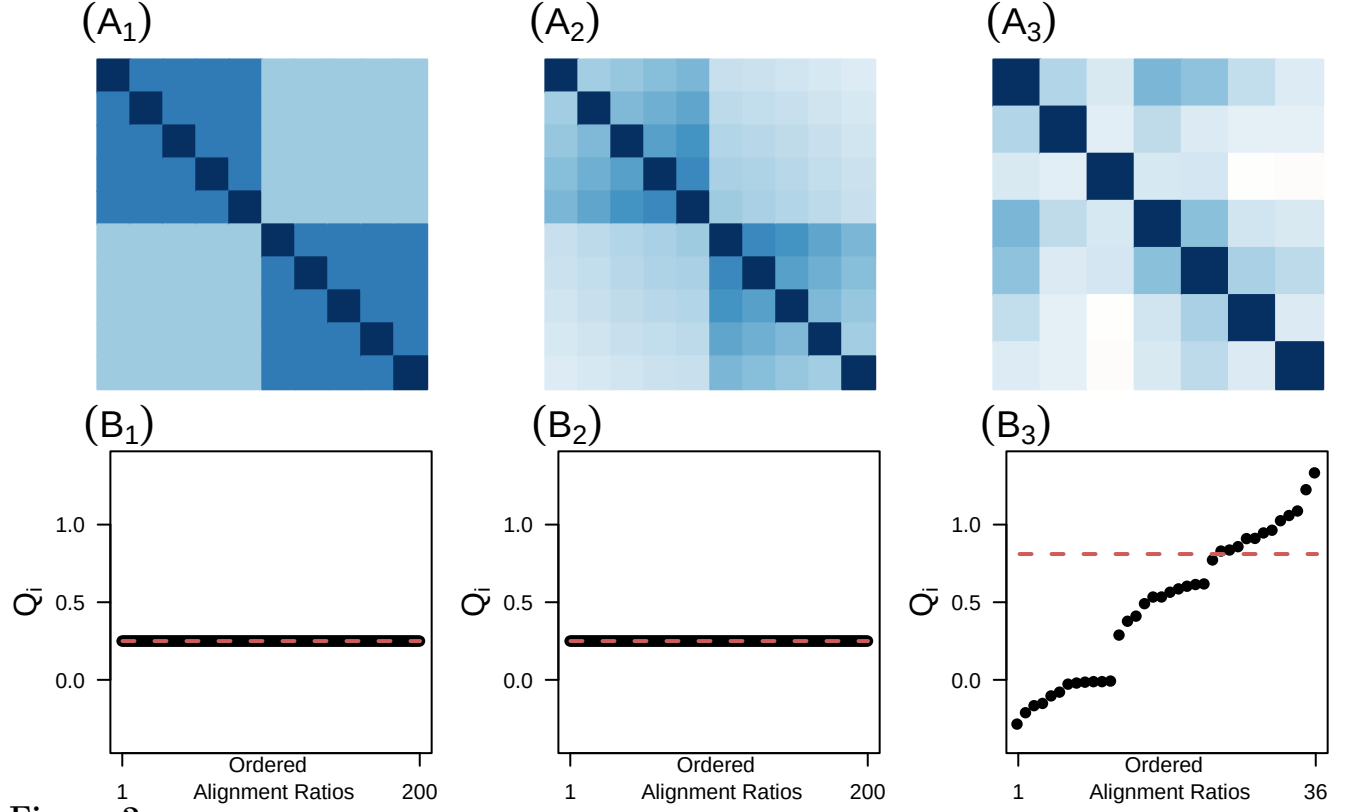
For convenience, we present the correction for attenuation on the correlation scale, although the same derivation holds for covariances. We therefore standardize the observed variables, $\text{var}(y_{ij}) = 1$. Under essential parallelism, tasks within each cluster have the same partition of variance into common and residual components. When two such clusters are specified within an ICM-CFA, the pairwise task correlations are constant within each cluster and between the two clusters:

$$\begin{aligned} \text{cor}(y_{i\ell}, y_{i\ell'}) &= \lambda_A^2, & \ell \neq \ell' \in \mathcal{A}, \\ \text{cor}(y_{ik}, y_{ik'}) &= \lambda_B^2, & k \neq k' \in \mathcal{B}, \\ \text{cor}(y_{i\ell}, y_{ik}) &= \lambda_A \lambda_B \varphi, & \ell \in \mathcal{A}, k \in \mathcal{B}. \end{aligned}$$

Under these constraints, the population manifest correlation matrix therefore has block compound symmetry:

$$\boldsymbol{\rho} = \begin{bmatrix} (1 - \rho_A)\mathbf{I}_{m_A} + \rho_A \mathbf{1}_{m_A \times m_A} & \rho_{AB} \mathbf{1}_{m_A \times m_B} \\ \rho_{AB} \mathbf{1}_{m_B \times m_A} & (1 - \rho_B)\mathbf{I}_{m_B} + \rho_B \mathbf{1}_{m_B \times m_B} \end{bmatrix},$$

where $\mathbf{I}_d$ is the $d \times d$ identity matrix and $\mathbf{1}_{a \times b}$ is an $a \times b$ matrix of ones. The quantities ρ_A and ρ_B denote the common within-cluster correlations in A and B , respectively, and ρ_{AB} denotes the common between-cluster correlation. Figure 2A₁ shows an example of this block compound-symmetric correlation structure.

**Figure 2**

Correlation patterns for three cases. (A₁)–(A₃) show manifest correlation matrices, and (B₁)–(B₃) show the corresponding ordered alignment ratios. Columns 1 and 2 are synthetic aligned cases with the same target latent correlation: Column 1 satisfies essentially parallel measurement, whereas Column 2 is an aligned congeneric case with unequal loadings. Column 3 uses Tsukahara et al. (2020) data restricted to two clusters (attentional control as the first three tasks and sensory discrimination as the last four tasks, highlighted in Figure 1) and shows misalignment through non-invariant alignment ratios. The dashed red line marks the ICM-CFA fitted value of $\hat{\varphi}^2$. ICM-CFA models were fit in R using lavaan (Rosseel, 2012).

From the block compound-symmetric structure, the following equalities hold:

$$\lambda_A^2 = \rho_A, \quad \lambda_B^2 = \rho_B, \quad \lambda_A \lambda_B \varphi = \rho_{AB}.$$

With the latent factors oriented so that $\lambda_A, \lambda_B > 0$, these equalities imply

$$\varphi = \frac{\rho_{AB}}{\sqrt{\rho_A \rho_B}}, \quad |\rho_{AB}| \leq \sqrt{\rho_A \rho_B},$$

where the inequality follows from $|\varphi| \leq 1$. The identity can also be written on the covariance scale.¹

The interpretation of φ is now transparent. The expression is Spearman’s correction for attenuation (Revelle & Condon, 2019; Spearman, 1904). Under essentially parallel measurement, the within-cluster correlations ρ_A and ρ_B are the reliabilities of the individual measures, while ρ_{AB} is their observed correlation. The latent correlation φ is therefore the correlation between their error-free components, or equivalently, the observed correlation corrected for measurement error.

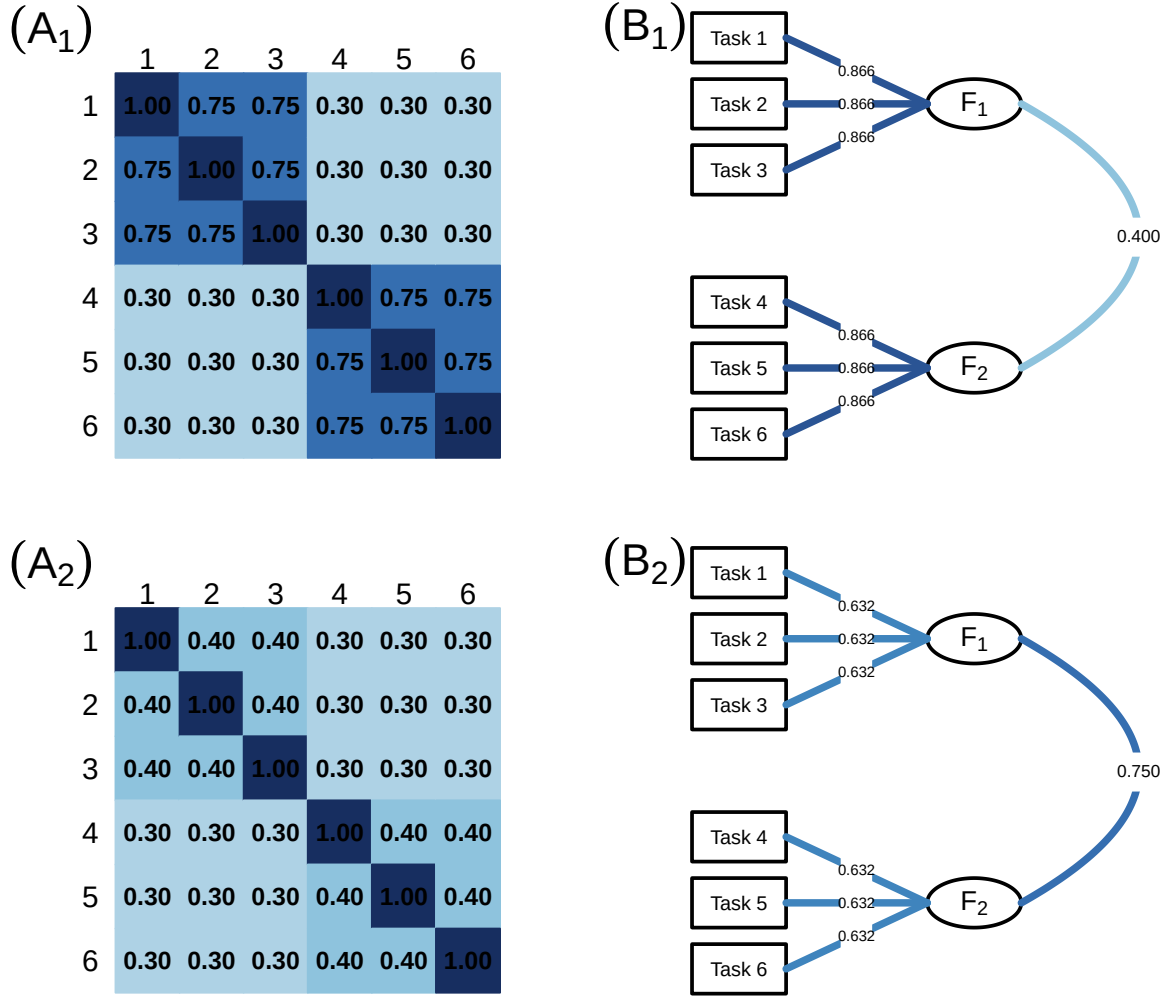
Figure 3 shows the consequence. Across rows, the between-cluster correlations are held fixed while the within-cluster correlations decrease, and the fitted latent correlation $\hat{\varphi}$ increases sharply. Low reliability therefore provides one explanation for large latent correlations amid low manifest correlations. This interpretation, however, depends on the restrictive measurement structure assumed here and does not extend to ICM-CFA generally.

Alignment: Ratios and Invariance

The essentially parallel structure considered above is restrictive. We now allow the indicators within each cluster to be congeneric, while retaining the broader ICM-CFA structure. Loadings and residual variances may therefore differ across tasks, so the within-cluster and between-cluster manifest associations need not be constant. Even so, the ICM-CFA imposes algebraic constraints linking the within-cluster and between-cluster covariance submatrices. These constraints are the focus of this section.

A classic example of how model structure constrains manifest covariances is vanishing tetrads (Bollen & Ting, 1993, 1998; Drton et al., 2007; Hipp & Bollen, 2003). Under a one-factor model, the common part of the covariance matrix is rank one, implying exact

¹ On the covariance scale, with standardized latent variables and $\lambda_A, \lambda_B > 0$, $\sigma_A = \lambda_A^2$, $\sigma_B = \lambda_B^2$, and $\sigma_{AB} = \lambda_A \lambda_B \varphi$, so $\varphi = \sigma_{AB} / \sqrt{\sigma_A \sigma_B}$.

**Figure 3**

Demonstration of latent-correlation behavior under essentially parallel measurement. (A₁) and (A₂) show two cluster-structured correlation matrices with identical between-cluster correlations but different within-cluster correlations. (B₁) and (B₂) show the corresponding two-factor ICM-CFA solutions. With the between-cluster correlations fixed, the fitted latent correlation increases as the within-cluster correlations decrease, consistent with Spearman disattenuation. ICM-CFA models were fit in R using lavaan (Rosseel, 2012).

equalities among certain products of off-diagonal covariances that can be tested.

Analogously, ICM-CFA implies that, for any pair of clusters, the corresponding

between-cluster covariance submatrix is rank one. However, rank one is necessary but not sufficient for ICM-CFA. ICM-CFA implies a stronger constraint, which we call *alignment*, and rank one follows as a consequence.

An illustration of alignment is shown in Figure 2A₂. When within-cluster loadings vary, they induce a systematic pattern in the between-cluster covariances. Tasks with larger loadings within a cluster tend to show larger covariances with tasks in the other cluster. This pattern can be expressed as invariances of certain covariance ratios, and the existence of a single latent correlation across clusters implies these invariances. We next define alignment formally and prove the implied constraints.

The alignment-ratio constraint

The development is most convenient in the context of two clusters, and the result is generalized subsequently. Consider J tasks partitioned into two clusters A and B , with sizes m_A and m_B (such that $m_A + m_B = J$). Without loss of generality, order the tasks so that the first m_A tasks belong to A and the remaining m_B belong to B . Define the corresponding index sets as

$$\mathcal{A} = \{1, \dots, m_A\}, \quad \mathcal{B} = \{m_A + 1, \dots, J\}.$$

Let λ_j denote the loading of task j on its cluster factor. Collect these loadings into the vectors $\boldsymbol{\lambda}_{\mathcal{A}} = (\lambda_1, \dots, \lambda_{m_A})'$ and $\boldsymbol{\lambda}_{\mathcal{B}} = (\lambda_{m_A+1}, \dots, \lambda_J)'$. Under ICM-CFA, each task loads on one and only one factor, so the $J \times 2$ loading matrix has the block form

$$\boldsymbol{\Lambda} = \begin{bmatrix} \boldsymbol{\lambda}_{\mathcal{A}} & \mathbf{0}_{m_A} \\ \mathbf{0}_{m_B} & \boldsymbol{\lambda}_{\mathcal{B}} \end{bmatrix}, \tag{1}$$

where $\mathbf{0}_d$ is a column vector of zeros of length d .

The manifest covariance matrix $\boldsymbol{\Sigma}$ can be partitioned into the two within-cluster

blocks and the between-cluster off-diagonal blocks as

$$\Sigma = \Lambda \Phi \Lambda' + \Psi = \begin{bmatrix} \Sigma_{AA} & \Sigma_{AB} \\ \Sigma'_{AB} & \Sigma_{BB} \end{bmatrix}. \quad (2)$$

We use the identification $\text{diag}(\Phi) = \mathbf{1}_2$, so $\Phi = \begin{bmatrix} 1 & \varphi \\ \varphi & 1 \end{bmatrix}$.

The two within-cluster blocks depend only on their respective loading vectors and residual variances:

$$\Sigma_{AA} = \lambda_A \lambda_A' + \Psi_A, \quad \Sigma_{BB} = \lambda_B \lambda_B' + \Psi_B,$$

where $\Psi_A = \text{diag}(\psi_1, \dots, \psi_{m_A})$ and $\Psi_B = \text{diag}(\psi_{m_A+1}, \dots, \psi_J)$. The between-cluster block depends on both loading vectors and is scaled by the latent correlation φ :

$$\Sigma_{AB} = \varphi \lambda_A \lambda_B'. \quad (3)$$

Because Σ_{AB} is an outer product, its rank is at most one. This rank-one condition is necessary in the two-cluster ICM-CFA case, but it is not the main restriction. The stronger restriction is that the same loading vectors λ_A and λ_B determine both the within-cluster and between-cluster covariances, which induces an elementwise alignment between them. To see this, we work with the off-diagonal entries of Σ , which do not depend on the residual matrix Ψ . The within-cluster and between-cluster covariance terms are as follows:

$$\begin{aligned} \Sigma_{\ell, \ell'} &= \lambda_\ell \lambda_{\ell'}, \quad \ell \neq \ell' \in \mathcal{A}, \\ \Sigma_{k, k'} &= \lambda_k \lambda_{k'}, \quad k \neq k' \in \mathcal{B}, \\ \Sigma_{\ell, k} &= \varphi \lambda_\ell \lambda_k, \quad \ell \in \mathcal{A}, k \in \mathcal{B}. \end{aligned} \quad (4)$$

The first two identities give within-cluster covariance terms from Σ_{AA} and Σ_{BB} , respectively, and the last identity gives a between-cluster covariance term from Σ_{AB} . Throughout, we use indices ℓ, ℓ' for tasks in cluster A and k, k' for tasks in cluster B . Using

these identities, the alignment property may be stated as

$$\begin{aligned}\Sigma_{\ell,k}\Sigma_{\ell',k'} &= (\varphi\lambda_\ell\lambda_k)(\varphi\lambda_{\ell'}\lambda_{k'}) \\ &= \varphi^2(\lambda_\ell\lambda_{\ell'})(\lambda_k\lambda_{k'}) \\ &= \varphi^2\Sigma_{\ell,\ell'} \cdot \Sigma_{k,k'}.\end{aligned}$$

Rearranging these identities gives the squared latent correlation in terms of covariance entries:

$$\varphi^2 = \frac{\Sigma_{\ell,k}\Sigma_{\ell',k'}}{\Sigma_{\ell,\ell'}\Sigma_{k,k'}}, \quad \Sigma_{\ell,\ell'} \neq 0, \Sigma_{k,k'} \neq 0. \quad (5)$$

To interpret Equation (5), define Q as its right-hand side:

$$Q(\ell, \ell', k, k') = \frac{\Sigma_{\ell,k} \Sigma_{\ell',k'}}{\Sigma_{\ell,\ell'} \Sigma_{k,k'}}. \quad (6)$$

We call Q an *alignment ratio* because it describes, at the level of covariance entries, how the between-cluster block aligns with the within-cluster blocks. The defining constraint of ICM-CFA is that these alignment ratios are invariant across index choices. For all admissible quadruples (ℓ, ℓ', k, k') and $(\tilde{\ell}, \tilde{\ell}', \tilde{k}, \tilde{k}')$, the following holds:

$$Q(\ell, \ell', k, k') = \varphi^2 = Q(\tilde{\ell}, \tilde{\ell}', \tilde{k}, \tilde{k}'). \quad (7)$$

The invariance of alignment ratios is the defining constraint implied by ICM-CFA. It implies that the between-cluster covariances are linked to within-cluster covariances through entrywise ratio constraints, so that every admissible quadruple (ℓ, ℓ', k, k') identifies the same value φ^2 . Consequently, φ is not the disattenuation of a single manifest correlation. Rather, φ^2 is the invariant value linking the between-cluster and within-cluster covariances. While interpreting the latent correlation as a disattenuated correlation is strained, its interpretation as a correlation between two latent factors remains. Equation (7) not only defines the invariance of alignment ratios, it also implies the constraint $0 \leq Q(\ell, \ell', k, k') \leq 1$, since $Q(\ell, \ell', k, k') = \varphi^2$ and $|\varphi| \leq 1$.

Alignment ratios applied

One target of analysis is to assess how alignment ratios vary across admissible quadruples. The bottom row of Figure 2 shows these ratios, sorted from smallest to largest, for covariance matrices that obey ICM-CFA (Panels B₁ and B₂) and for a covariance matrix that does not (Panel B₃). The invariant value is represented as a horizontal line; violations appear as dispersion around it. Under ICM-CFA, the common value equals φ^2 .

How many admissible alignment ratios are there? An admissible alignment ratio requires two distinct indices $\ell \neq \ell'$ from cluster $\mathcal{A}$ and two distinct indices $k \neq k'$ from cluster $\mathcal{B}$, with $\Sigma_{\ell\ell'}\Sigma_{kk'} \neq 0$ so the denominator is defined. Because Σ is symmetric, the within-cluster order is immaterial, so we treat the selections as unordered pairs $\{\ell, \ell'\} \subset \mathcal{A}$ and $\{k, k'\} \subset \mathcal{B}$. For any fixed choice of these two unordered pairs, there are exactly two distinct between-cluster matchings in the numerator of Equation (6). In one matching, ℓ is paired with k and ℓ' with k' (yielding $\Sigma_{\ell k}\Sigma_{\ell' k'}$); in the other, ℓ is paired with k' and ℓ' with k (yielding $\Sigma_{\ell k'}\Sigma_{\ell' k}$). We indicate which matching is used by the order of the $\mathcal{B}$ indices in the arguments of Q :

$$Q(\ell, \ell', k, k') = \frac{\Sigma_{\ell k}\Sigma_{\ell' k'}}{\Sigma_{\ell\ell'}\Sigma_{kk'}} \quad \text{and} \quad Q(\ell, \ell', k', k) = \frac{\Sigma_{\ell k'}\Sigma_{\ell' k}}{\Sigma_{\ell\ell'}\Sigma_{kk'}}.$$

The two cross-matchings are distinct, whereas within-cluster swaps merely exchange or relabel these two ratios (e.g., $Q(\ell', \ell, k, k') = Q(\ell, \ell', k', k)$ and $Q(\ell', \ell, k', k) = Q(\ell, \ell', k, k')$). Thus each unordered pair $\{\ell, \ell'\}$ and $\{k, k'\}$ yields exactly two distinct alignment ratios. It follows that a cluster pair with m_A and m_B tasks has

$$n_Q(m_A, m_B) = 2 \binom{m_A}{2} \binom{m_B}{2}$$

distinct alignment ratios.

Pairwise locality

The two-cluster result extends directly to any number of clusters. Suppose the tasks are partitioned into K clusters, $\mathcal{C}_1, \dots, \mathcal{C}_K$, with latent correlation matrix $\Phi = (\varphi_{XY})$. For

any pair of clusters $\mathcal{C}_X$ and $\mathcal{C}_Y$, admissible alignment ratios are defined exactly as above, using two distinct tasks from $\mathcal{C}_X$ and two distinct tasks from $\mathcal{C}_Y$. No three-cluster quadruples are involved.

Under ICM-CFA with K clusters, $\mathbf{\Lambda}$ is block structured by cluster, so the off-diagonal entries of $\mathbf{\Sigma}$ generalize Equation (4) as

$$\begin{aligned}\Sigma_{\ell,\ell'} &= \lambda_\ell \lambda_{\ell'}, & \ell \neq \ell' \in \mathcal{C}_X, \\ \Sigma_{\ell,k} &= \varphi_{XY} \lambda_\ell \lambda_k, & \ell \in \mathcal{C}_X, k \in \mathcal{C}_Y, X \neq Y.\end{aligned}$$

The within-cluster identity does not involve $\mathbf{\Phi}$ at all, and each between-cluster identity involves only the single entry φ_{XY} for the pair of clusters to which the two tasks belong.

For $\ell, \ell' \in \mathcal{C}_X$ with $\ell \neq \ell'$ and $k, k' \in \mathcal{C}_Y$ with $k \neq k'$, the ICM-CFA covariance structure gives

$$Q_{XY}(\ell, \ell', k, k') = \frac{\Sigma_{\ell k} \Sigma_{\ell' k'}}{\Sigma_{\ell \ell'} \Sigma_{k k'}} = \frac{(\varphi_{XY} \lambda_\ell \lambda_k)(\varphi_{XY} \lambda_{\ell'} \lambda_{k'})}{(\lambda_\ell \lambda_{\ell'})(\lambda_k \lambda_{k'})} = \varphi_{XY}^2. \quad (8)$$

Thus, alignment-ratio invariance is local to pairs of clusters. Each pair (X, Y) has its own invariant value, φ_{XY}^2 , and the full K -cluster model imposes these constraints across the $\binom{K}{2}$ cluster pairs. The only remaining global constraint is that the pairwise latent correlations must form a valid positive semidefinite latent correlation matrix $\mathbf{\Phi}$. In particular, the invariance results of the preceding sections hold verbatim within each cluster pair. This locality is definitional rather than diagnostic. Certain misspecifications may affect several cluster pairs at once, but each alignment-ratio condition is defined strictly between two clusters.

Misspecification

In the preceding sections, we derived the invariance of alignment ratios as a defining consequence of ICM-CFA. In the next two sections, we study how violations of this invariance affect the estimation and interpretation of the ICM-CFA latent correlation. The two sections consider different types of violations. In the next section, data are generated

from an ICM-CFA model, but the analyst assigns one or more tasks to the wrong cluster. We refer to this misspecification as *misassignment*. In the following section, we assume the analyst assigns tasks to clusters correctly, but the within- and between-cluster submatrices nonetheless fail to align, so the alignment ratios are not invariant. We refer to this misspecification as *misalignment*.

For both types of violations, the main question is what happens to the latent correlation when an ICM-CFA model is fit. As discussed, previous work has argued that ICM-CFA misspecification can produce fitted latent correlations much larger than the corresponding manifest associations between clusters (Asparouhov & Muthén, 2009; Marsh et al., 2009; Marsh et al., 2014). We use the alignment-ratio framework to examine this claim. To foreshadow, the pattern is evident under both misassignment and misalignment.

Misspecification Through Misassignment

In this section, we analyze the effect of misspecifying the partition of tasks into clusters. This type of misspecification is common, particularly in behavioral experiments where the true grouping is rarely known with certainty. In practice, tasks are often grouped by face validity, labels, or precedent (Brown, 2015), and misassignments are likely when data are noisy and cluster structure is less salient. Such misspecification can distort parameter estimates, including latent correlations (Asparouhov & Muthén, 2009; Marsh et al., 2009). When the assumed assignment is incorrect, some covariance terms that are truly within-cluster are treated as between-cluster, and some truly between-cluster terms are treated as within-cluster. This reclassification increases the apparent between-cluster association and decreases within-cluster cohesion. As we show below, in the two-cluster case, no alignment ratio falls below its correctly specified value. This one-sided distortion is consistent with inflated fitted latent correlations.

Consider the case where J tasks are divided into two true clusters, A and B , each of size m (so $2m = J$). Suppose an analyst misassigns one task from cluster A to cluster B ,

and vice versa. Let $\mathcal{A}$ and $\mathcal{B}$ denote the true index sets for the two clusters, such that $\mathcal{A} = \{1, \dots, m\}$ and $\mathcal{B} = \{m+1, \dots, 2m\}$. Let $\hat{\mathcal{A}}$ and $\hat{\mathcal{B}}$ denote the analyst's assumed index sets, such that each contains exactly one misassigned task. For any admissible within-cluster pairs $\ell \neq \ell' \in \hat{\mathcal{A}}$ and $k \neq k' \in \hat{\mathcal{B}}$, each pair is either affected or unaffected by the misassignment. As a result, each admissible quadruple (ℓ, ℓ', k, k') that forms an alignment ratio falls into one of three cases: (i) both within-cluster pairs are unaffected, (ii) exactly one is affected, or (iii) both are affected.

Given a generative ICM-CFA model with a true latent correlation φ , we analyze the alignment ratios that arise under the misassigned partition for each of the three cases.

Case i. Both assumed within-cluster pairs are unaffected: $\ell \neq \ell' \in \hat{\mathcal{A}} \cap \mathcal{A}$ and $k \neq k' \in \hat{\mathcal{B}} \cap \mathcal{B}$. The alignment ratio remains at the correctly labeled value φ^2 . Recall that the alignment ratio $Q(\ell, \ell', k, k')$ can be defined from entries of the manifest covariance matrix as

$$Q(\ell, \ell', k, k') = \frac{\Sigma_{\ell, k} \Sigma_{\ell', k'}}{\Sigma_{\ell, \ell'} \Sigma_{k, k'}}.$$

We can use Equation (4) to define each of the covariance terms in terms of model parameters as

$$Q(\ell, \ell', k, k') = \frac{(\varphi \lambda_\ell \lambda_k)(\varphi \lambda_{\ell'} \lambda_{k'})}{(\lambda_\ell \lambda_{\ell'})(\lambda_k \lambda_{k'})} = \varphi^2 \leq 1$$

Given that there are $m-1$ unaffected tasks in each assumed cluster, the number of admissible index quadruples (ℓ, ℓ', k, k') that fall into Case i is $\binom{m-1}{2}^2$. Each such quadruple yields two admissible alignment ratios (corresponding to the two between-cluster matchings), so the number of Case i alignment ratios is

$$N_1 = 2 \binom{m-1}{2}^2.$$

Therefore, out of the $2 \binom{m}{2}^2$ admissible alignment ratios overall, N_1 equal the true value φ^2 .

Case ii. Exactly one assumed within-cluster pair is affected. Without loss of

generality, suppose $\ell^* \in \hat{\mathcal{A}} \cap \mathcal{B}$ is the misassigned index in $\hat{\mathcal{A}}$, and $\ell' \in \hat{\mathcal{A}} \cap \mathcal{A}$ is correctly assigned. Assume the within- $\hat{\mathcal{B}}$ pair is unaffected, $k \neq k' \in \hat{\mathcal{B}} \cap \mathcal{B}$. Then $\Sigma_{\ell^*,k}$ is, in truth, a within- $\mathcal{B}$ covariance term, whereas $\Sigma_{\ell^*,\ell'}$ is, in truth, a between-cluster covariance term. The remaining two terms fall in their correct cluster positions. Defining the alignment ratio for this case gives

$$\begin{aligned} Q(\ell^*, \ell', k, k') &= \frac{\Sigma_{\ell^*,k} \Sigma_{\ell',k'}}{\Sigma_{\ell^*,\ell'} \Sigma_{k,k'}} \\ &= \frac{(\lambda_{\ell^*} \lambda_k)(\varphi \lambda_{\ell'} \lambda_{k'})}{(\varphi \lambda_{\ell^*} \lambda_{\ell'})(\lambda_k \lambda_{k'})} = 1. \end{aligned}$$

Since ℓ^* may be paired with any of the $m - 1$ unaffected tasks in $\hat{\mathcal{A}}$, and (k, k') may be any of the $\binom{m-1}{2}$ unaffected pairs in $\hat{\mathcal{B}}$, there are $(m - 1)\binom{m-1}{2}$ such index quadruples in this subcase. By symmetry, the same count holds when the affected pair is in $\hat{\mathcal{B}}$, giving $2(m - 1)\binom{m-1}{2}$ Case ii index quadruples overall. Each such quadruple yields two admissible alignment ratios, and in this case both equal 1. Hence the total number of Case ii alignment ratios is

$$N_2 = 4(m - 1)\binom{m - 1}{2}.$$

Thus, N_2 alignment ratios equal 1 in Case ii.

Case iii. Both assumed within-cluster pairs are affected. Let $\ell^* \in \hat{\mathcal{A}} \cap \mathcal{B}$ be the misassigned index in $\hat{\mathcal{A}}$ (truly in $\mathcal{B}$), and let $k^* \in \hat{\mathcal{B}} \cap \mathcal{A}$ be the misassigned index in $\hat{\mathcal{B}}$ (truly in $\mathcal{A}$). Choose corresponding unaffected indices $\ell \in \hat{\mathcal{A}} \cap \mathcal{A}$ and $k \in \hat{\mathcal{B}} \cap \mathcal{B}$. For each choice, there are two possible matchings across the numerator.

If the numerator pairs ℓ^* with k^* and ℓ with k , then all four terms correspond to between-cluster covariances and the ratio collapses to

$$\begin{aligned} Q(\ell^*, \ell, k^*, k) &= \frac{\Sigma_{\ell^*,k^*} \Sigma_{\ell,k}}{\Sigma_{\ell^*,\ell} \Sigma_{k^*,k}} \\ &= \frac{(\varphi \lambda_{\ell^*} \lambda_{k^*})(\varphi \lambda_{\ell} \lambda_k)}{(\varphi \lambda_{\ell^*} \lambda_{\ell})(\varphi \lambda_{k^*} \lambda_k)} = 1. \end{aligned}$$

If instead the numerator pairs ℓ^* with k and ℓ with k^* , then both numerator terms

are true within-cluster covariances while both denominator terms are true between-cluster covariances, giving

$$\begin{aligned} Q(\ell^*, \ell, k, k^*) &= \frac{\Sigma_{\ell^*, k} \Sigma_{\ell, k^*}}{\Sigma_{\ell^*, \ell} \Sigma_{k^*, k}} \\ &= \frac{(\lambda_{\ell^*} \lambda_k)(\lambda_{\ell} \lambda_{k^*})}{(\varphi \lambda_{\ell^*} \lambda_{\ell})(\varphi \lambda_{k^*} \lambda_k)} = \frac{1}{\varphi^2} \geq 1. \end{aligned}$$

Given that there are $m - 1$ choices for ℓ and $m - 1$ choices for k , there are $(m - 1)^2$ underlying index quadruples in Case iii. Since each yields two admissible matchings, the total number of Case iii alignment ratios is

$$N_3 = 2(m - 1)^2.$$

In this setting, each choice of (ℓ, k) yields two alignment ratios, one equal to 1 and one equal to $1/\varphi^2$. Consequently, across all Case iii ratios, half equal 1 and half equal $1/\varphi^2$.

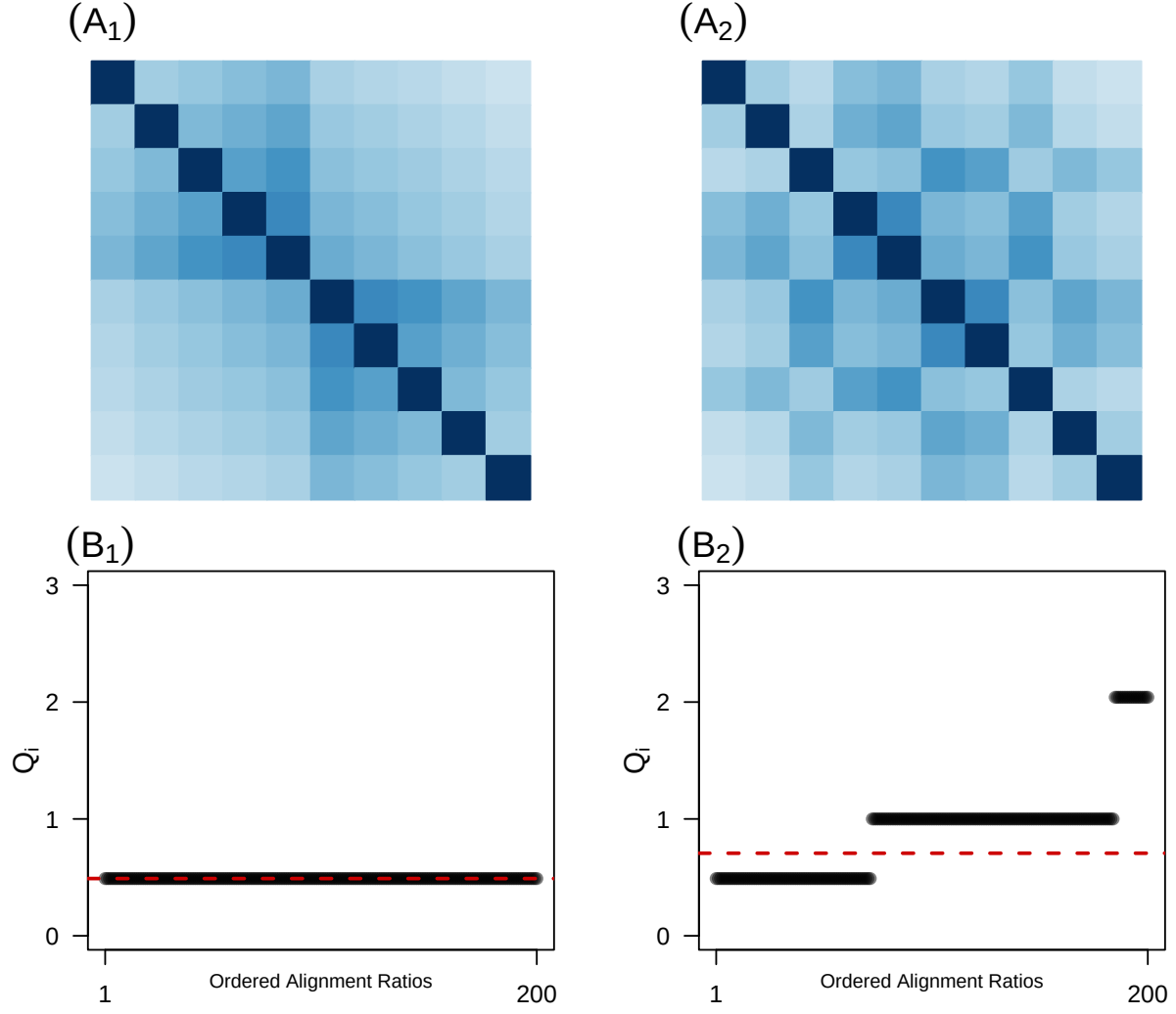
The alignment ratios are no longer invariant under misassignment. Instead, the admissible ratios take three values $(\varphi^2, 1, 1/\varphi^2)$ with frequencies $(N_1, N_2 + \frac{N_3}{2}, \frac{N_3}{2})$. Only the first value equals the correctly labeled target φ^2 ; all remaining ratios equal 1 or exceed 1.

Figure 4 illustrates this pattern. The top row shows the correlation matrix under the correctly labeled and task-swapped assignments (Panels A₁ and A₂). Panel B₁ shows the correctly labeled case, where all alignment ratios coincide. Panel B₂ shows the misassigned case, where the ordered ratios separate into three plateaus corresponding to the three values above.

General misassignment patterns

The three-value pattern is not specific to the symmetric single-swap example. It holds more generally for any way of misassigning tasks between two clusters. The assumed clusters may have equal or unequal sizes, and the errors may occur in one or both assumed clusters.

To show this generality, suppose the data are generated by a two-cluster ICM-CFA with true clusters A and B , indexed by $\mathcal{A}$ and $\mathcal{B}$, respectively. Let $z_j = 0$ if task j truly

**Figure 4**

Misassignment breaks alignment-ratio invariance. (A₁) shows the correctly labeled correlation matrix and (A₂) the matrix after swapping one task across clusters. (B₁) shows that ordered alignment ratios are invariant, whereas (B₂) shows three plateaus at φ^2 , 1, and $1/\varphi^2$. The dashed line marks the fitted $\hat{\varphi}^2$.

belongs to $\mathcal{A}$, and let $z_j = 1$ if task j truly belongs to $\mathcal{B}$. Then each off-diagonal covariance can be written as

$$\Sigma_{j,j'} = \begin{cases} \lambda_j \lambda_{j'}, & z_j = z_{j'}, \\ \varphi \lambda_j \lambda_{j'}, & z_j \neq z_{j'}. \end{cases}$$

It is helpful to define the indicator function that is zero when two tasks are in the same cluster and one otherwise:

$$e(j, j') = \mathbb{I}(z_j \neq z_{j'}) = |z_j - z_{j'}|.$$

With this indicator, $\Sigma_{j,j'} = \varphi^{e(j,j')} \lambda_j \lambda_{j'}$.

Now take any assumed partition $(\hat{\mathcal{A}}, \hat{\mathcal{B}})$, even if it is wrong. For any admissible quadruple $\ell \neq \ell' \in \hat{\mathcal{A}}$ and $k \neq k' \in \hat{\mathcal{B}}$, the alignment ratio is

$$Q(\ell, \ell', k, k') = \frac{\Sigma_{\ell k} \Sigma_{\ell' k'}}{\Sigma_{\ell \ell'} \Sigma_{k k'}}.$$

Substituting the two-cluster covariance form gives

$$Q(\ell, \ell', k, k') = \varphi^d,$$

where

$$d = e(\ell, k) + e(\ell', k') - e(\ell, \ell') - e(k, k').$$

The loadings cancel because each loading appears once in the numerator and once in the denominator. Thus, the ratio depends only on how the four tasks are distributed across the true clusters. Because $z_j \in \{0, 1\}$, $e(j, j') = z_j + z_{j'} - 2z_j z_{j'}$, which implies that $d = 2(z_\ell - z_{k'})(z_{\ell'} - z_k) \in \{-2, 0, 2\}$. Therefore, every alignment ratio under any assumed two-cluster partition must take one of three values:

$$Q(\ell, \ell', k, k') \in \left\{ \varphi^2, 1, \frac{1}{\varphi^2} \right\}.$$

These three values also imply a one-sided ordering. Assume that all loadings are nonzero, $0 < |\varphi| < 1$, and each true and assumed cluster contains at least two tasks; the clusters need not be equal in size. The three values then satisfy $\varphi^2 < 1 < 1/\varphi^2$. Because every ratio under the correct partition equals φ^2 , no ratio under an assumed partition can fall below it. To establish strict dominance, it remains to show that every genuinely incorrect partition produces at least one larger ratio.

If the assumed partition is neither correct nor a relabeling, at least one assumed cluster contains tasks from both true clusters. Select one task from each true cluster within that assumed cluster and any admissible pair from the other assumed cluster. Across the two possible matchings, at least one has $d = 0$ or $d = -2$, producing a ratio of 1 or $1/\varphi^2$. Thus, every nontrivial misassignment leaves all ratios at or above φ^2 and moves at least one above it.

When the correct and assumed partitions generate the same number N of ratios, let $Q_{(r)}(\mathcal{A}, \mathcal{B})$ and $Q_{(r)}(\hat{\mathcal{A}}, \hat{\mathcal{B}})$ denote the r th smallest ratios under the true and assumed partitions, respectively. Then

$$Q_{(r)}(\hat{\mathcal{A}}, \hat{\mathcal{B}}) \geq Q_{(r)}(\mathcal{A}, \mathcal{B}) = \varphi^2, \quad r = 1, \dots, N,$$

with strict inequality for at least one r . It follows that any summary that is nondecreasing in the ordered ratios cannot be smaller under the assumed partition; if it is strictly increasing in each argument, it must be larger. If the two collections have different numbers of ratios, an entrywise comparison is not available, but every ratio under the assumed partition remains bounded below by φ^2 , with at least one strictly larger. The geometric mean is strictly increasing in each argument and needs no entrywise comparison, so it exceeds φ^2 in both cases; the median is nondecreasing but not strictly increasing, so it cannot fall below φ^2 but need not exceed it.

Generalization to three or more clusters

The above one-sided result continues to hold locally for the pair containing a task's true and assumed clusters, but it does not extend across the full set of cluster pairs. With three or more clusters, a misassigned task can enter ratios involving several different latent correlations at once. For example, suppose a task ℓ^* truly belongs to cluster A but is assigned to cluster C . Then, for correctly assigned $k \neq k' \in \mathcal{B}$ and $c \in \mathcal{C}$, a ratio in the assumed (B, C) block gives

$$Q = \frac{\Sigma_{k, \ell^*} \Sigma_{k', c}}{\Sigma_{k, k'} \Sigma_{\ell^*, c}} = \frac{(\varphi_{AB} \lambda_k \lambda_{\ell^*})(\varphi_{BC} \lambda_{k'} \lambda_c)}{(\lambda_k \lambda_{k'})(\varphi_{AC} \lambda_{\ell^*} \lambda_c)} = \frac{\varphi_{AB} \varphi_{BC}}{\varphi_{AC}}.$$

This value is no longer restricted to $\{\varphi^2, 1, 1/\varphi^2\}$. It may even fall below the correctly specified pairwise value. For example, take $\varphi_{AB} = 0.5$ and $\varphi_{AC} = \varphi_{BC} = 0.8$.² In this case,

$$\frac{\varphi_{AB}\varphi_{BC}}{\varphi_{AC}} = 0.5 < 0.64 = \varphi_{BC}^2.$$

Thus, the one-sided pattern for misassignment is a two-cluster result. With three or more clusters, the effect of misassignment depends on the full pattern of latent correlations. Cluster pairs that do not contain misassigned tasks remain invariant at their pairwise values. A more general treatment of multi-cluster misassignment is beyond the scope of this paper.

Misspecification Through Misalignment

In the previous section, we considered misspecifications where tasks are misassigned to clusters. Here, we consider a different type of misspecification, one in which cluster assignments are correct but the alignment invariance required by ICM-CFA does not hold. ICM-CFA alignment invariance implies that between-cluster covariances have a constrained relationship to within-cluster covariances. When this alignment fails, ICM-CFA is misspecified even with correct cluster membership.

One convenient approach to generating misspecification by misalignment is to use the bifactor model. In the bifactor model, there is a general factor as well as cluster-specific factors (Holzinger & Swineford, 1937; Reise, 2012). A graphical depiction is shown in Figure 5A₁. We assume that the clusters are correctly assigned in the following analysis. A key feature of the bifactor model is that the general factor and all specific factors are orthogonal. Correlations between tasks in different clusters reflect loadings on the general factor alone, whereas correlations between tasks in the same cluster reflect contributions from both the general factor and the relevant specific factor. Alignment therefore holds only under special loading configurations. Unlike misassignment, misalignment does not reduce to a finite set of cases because the alignment ratios vary continuously with the general and specific loadings.

² The resulting Φ is positive definite, with determinant .11, and is therefore a valid latent correlation matrix.

We therefore first derive the conditions under which a bifactor model satisfies alignment and then study violations of these conditions numerically.

Alignment invariance for bifactor models

When are alignment ratios invariant in the bifactor model? The two-cluster case is sufficient to explore the question. We again have J tasks partitioned into clusters A and B , with index sets $\mathcal{A} = \{1, \dots, m\}$ and $\mathcal{B} = \{m+1, \dots, J\}$. Let γ_j denote the loading of task j on the general factor, let δ_j denote the loading of task j on its cluster-specific factor, and let ψ_j denote the unique variance of task j . Assume the general factor and the two cluster-specific factors are mutually orthogonal and standardized to variance 1. The population covariance matrix is

$$\mathbf{\Omega} = \boldsymbol{\gamma}\boldsymbol{\gamma}' + \mathbf{\Delta}\mathbf{\Delta}' + \mathbf{\Psi},$$

where $\boldsymbol{\gamma} = (\gamma_1, \dots, \gamma_J)'$ is the vector of general-factor loadings and $\mathbf{\Psi} = \text{diag}(\psi_1, \dots, \psi_J)$.

The cluster-specific loading matrix $\mathbf{\Delta}$ has block form

$$\mathbf{\Delta} = \begin{bmatrix} \boldsymbol{\delta}_{\mathcal{A}} & \mathbf{0}_{m \times 1} \\ \mathbf{0}_{(J-m) \times 1} & \boldsymbol{\delta}_{\mathcal{B}} \end{bmatrix},$$

with $\boldsymbol{\delta}_{\mathcal{A}} = (\delta_1, \dots, \delta_m)'$ and $\boldsymbol{\delta}_{\mathcal{B}} = (\delta_{m+1}, \dots, \delta_J)'$.

Using the same within- and between-cluster partition as in Equation (2), the off-diagonal entries of $\mathbf{\Omega}$ satisfy three identities analogous to Equation (4). The within-cluster and cross-cluster covariance terms are parameterized as

$$\begin{aligned} \Omega_{\ell, \ell'} &= \gamma_{\ell}\gamma_{\ell'} + \delta_{\ell}\delta_{\ell'}, & \ell \neq \ell' \in \mathcal{A}, \\ \Omega_{k, k'} &= \gamma_k\gamma_{k'} + \delta_k\delta_{k'}, & k \neq k' \in \mathcal{B}, \\ \Omega_{\ell, k} &= \gamma_{\ell}\gamma_k, & \ell \in \mathcal{A}, k \in \mathcal{B}. \end{aligned}$$

Here δ_j denotes the loading of task j on its cluster-specific factor, that is, the A -specific factor for $\ell \in \mathcal{A}$ and the B -specific factor for $k \in \mathcal{B}$.

The first two identities give within-cluster covariance terms, whereas the last identity gives between-cluster terms. The between-cluster covariances depend only on the general-factor loadings because the cluster-specific factors have disjoint support.

Consequently,

$$\mathbf{\Omega}_{\mathcal{AB}} = \boldsymbol{\gamma}_{\mathcal{A}} \boldsymbol{\gamma}'_{\mathcal{B}},$$

is rank one by construction. The issue is therefore not cluster membership or between-cluster rank, but whether the within-cluster and between-cluster covariances satisfy invariant alignment ratios. Alignment ratios are given by

$$Q(\ell, \ell', k, k') = \frac{\Omega_{\ell, k} \Omega_{\ell', k'}}{\Omega_{\ell, \ell'} \Omega_{k, k'}} \quad \ell \neq \ell' \in \mathcal{A}, k \neq k' \in \mathcal{B}.$$

Substituting the bifactor expressions for elements of $\mathbf{\Omega}$ yields:

$$\begin{aligned} Q(\ell, \ell', k, k') &= \frac{(\gamma_{\ell} \gamma_k)(\gamma_{\ell'} \gamma_{k'})}{(\gamma_{\ell} \gamma_{\ell'} + \delta_{\ell} \delta_{\ell'})(\gamma_k \gamma_{k'} + \delta_k \delta_{k'})} \\ &= \frac{(\gamma_{\ell} \gamma_{\ell'})(\gamma_k \gamma_{k'})}{(1 + \frac{\delta_{\ell} \delta_{\ell'}}{\gamma_{\ell} \gamma_{\ell'}})(1 + \frac{\delta_k \delta_{k'}}{\gamma_k \gamma_{k'}})(\gamma_{\ell} \gamma_{\ell'})(\gamma_k \gamma_{k'})} \\ &= \frac{1}{(1 + \frac{\delta_{\ell} \delta_{\ell'}}{\gamma_{\ell} \gamma_{\ell'}})(1 + \frac{\delta_k \delta_{k'}}{\gamma_k \gamma_{k'}})}. \end{aligned}$$

where the last equality assumes $\gamma_{\ell} \gamma_{\ell'} \neq 0$ and $\gamma_k \gamma_{k'} \neq 0$.

We are now in a position to ask what constraints on the loadings yield alignment-ratio invariance. To do so, it is helpful to define the *loading ratio*

$$\kappa_j = \frac{\delta_j}{\gamma_j}.$$

The loading ratio is task-specific and describes the strength of the cluster-specific loading relative to the general loading. Larger κ_j implies stronger within-cluster association relative to between-cluster association. The alignment ratio is

$$Q(\ell, \ell', k, k') = \frac{1}{(1 + \kappa_{\ell} \kappa_{\ell'})(1 + \kappa_k \kappa_{k'})}.$$

For alignment-ratio invariance, $Q(\ell, \ell', k, k')$ must be constant over all admissible quadruples with $\ell \neq \ell' \in \mathcal{A}$ and $k \neq k' \in \mathcal{B}$. This is possible only if, within each cluster, the pairwise products $\kappa_\ell \kappa'_{\ell'}$ and $\kappa_k \kappa'_{k'}$ do not depend on the choices of indices. This constraint leads immediately to

$$\begin{aligned}\kappa_\ell &= \alpha_A, & \forall \ell \in \mathcal{A}, \\ \kappa_k &= \alpha_B, & \forall k \in \mathcal{B}.\end{aligned}$$

Equivalently, invariance can be stated in terms of the loadings as

$$\begin{aligned}\delta_\ell &= \alpha_A \gamma_\ell, & \forall \ell \in \mathcal{A}, \\ \delta_k &= \alpha_B \gamma_k, & \forall k \in \mathcal{B}.\end{aligned}$$

This constraint is the *proportionality constraint* used by Gignac (2016), which characterizes when a higher-order model is nested within a bifactor model. Violating this constraint will be our method for studying misspecification by misalignment.

Linking ICM-CFA, Bifactor, and Hierarchical CFA Models

In general, ICM-CFA and the bifactor model are not nested within one another (Greene et al., 2019). They are linked through the hierarchical (second-order) CFA model, which is a special case of both. This hierarchical CFA is an ICM-CFA model when clusters share covariance through a single higher-order factor. Alternatively, this hierarchical CFA model may be reparameterized as a bifactor model via the Schmid-Leiman transformation (Brunner et al., 2012; Schmid & Leiman, 1957; Yung et al., 1999). The resulting bifactor is constrained—within each cluster the specific loadings are proportional to the general loadings (Gignac, 2016; Mansolf & Reise, 2017).

With two clusters, the ICM-CFA factor-correlation structure is automatically one-dimensional because there is only one cluster-factor correlation. Therefore, once within-cluster proportionality holds, the Schmid-Leiman decomposition yields a bifactor representation of the same covariance structure. Under proportionality, the ICM-CFA latent correlation has three complementary interpretations. First, it is a latent correlation between

the cluster factors. Second, it is a statement about alignment through invariant alignment ratios. And third, it indexes the relative strength of the general and specific components:

$$\varphi = \frac{1}{\sqrt{(1 + \alpha_A^2)(1 + \alpha_B^2)}}. \quad (9)$$

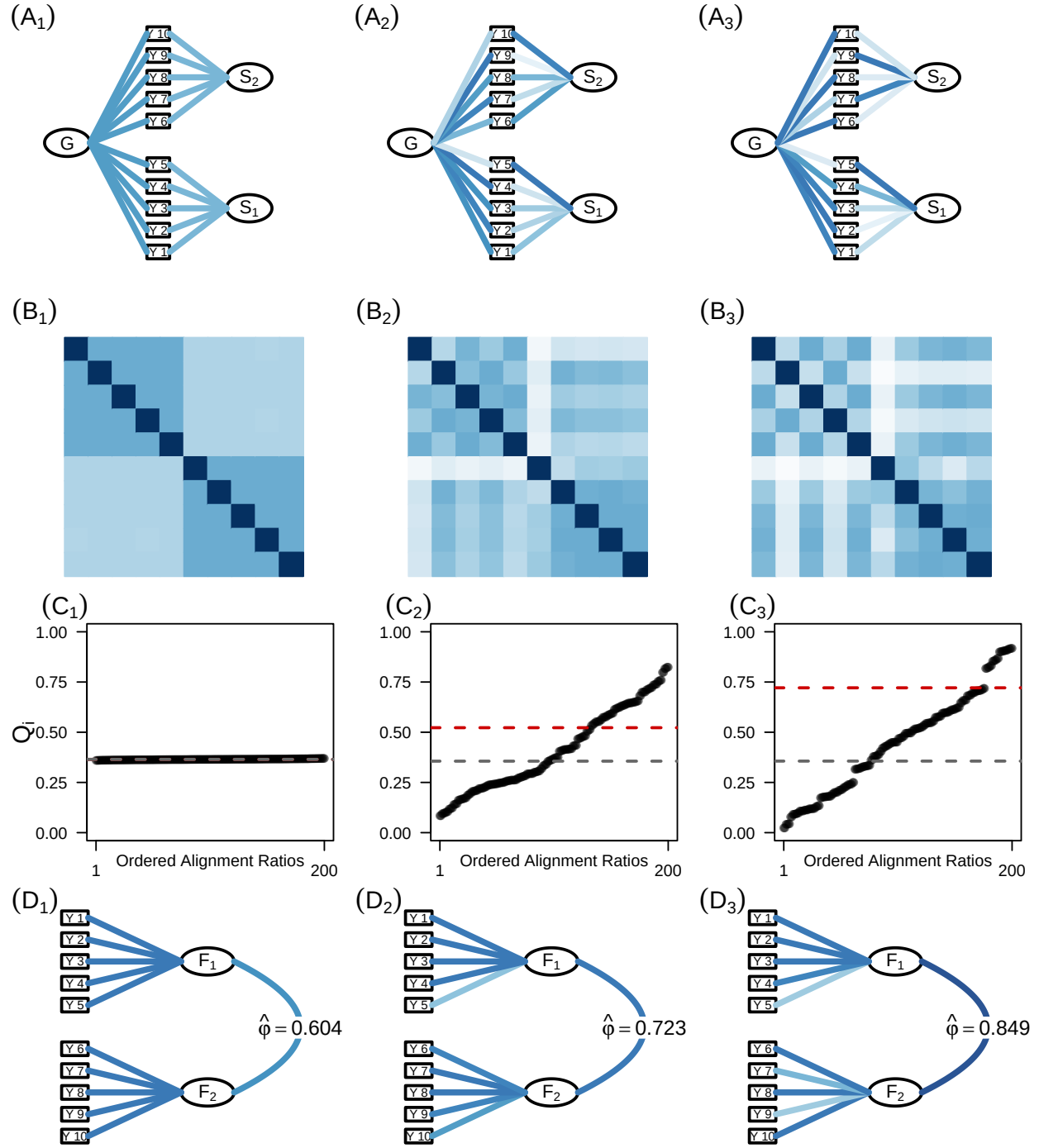
As α_A and α_B decrease (general loadings dominate specific loadings within each cluster), φ increases toward 1. Conversely, as α_A and α_B increase (specific loadings dominate general loadings within each cluster), φ decreases toward 0. Thus, under proportionality, φ indexes the extent to which the covariance shared across clusters is carried by a common general component rather than by cluster-specific components. This interpretation aligns with the use of latent correlations as a measure of discriminant validity (Brown, 2015; Campbell & Fiske, 1959; Fornell & Larcker, 1981; Henseler et al., 2015), with higher φ indicating lower discriminant validity and lower φ indicating higher discriminant validity.

Assessing the Effect of Misalignment

To assess the effect of misalignment on ICM-CFA, we use the bifactor model to generate manifest covariance matrices and then fit ICM-CFA to those matrices. When κ_j is constant within each cluster, the proportionality constraint holds and alignment invariance is satisfied, so the ICM-CFA model is correctly specified even though the data are generated from a bifactor model. In this case, the latent correlation is well defined and corresponds to the expression given above. The main question is what happens when κ_j varies across tasks within a cluster.

When κ_j varies within a cluster, the alignment ratios are no longer invariant across admissible quadruples and therefore do not imply a single population latent correlation. Instead, there are many different alignment ratios. We use the geometric mean of these population alignment ratios, denoted $\text{GM}(Q)$, as a central-tendency reference and compare it with the ICM-CFA fitted value $\hat{\varphi}^2$.

Figure 5 shows examples of the approach. The first column shows a bifactor model

**Figure 5**

Alignment under bifactor-generated data. Columns 1-3 vary the proportionality between general and specific loadings. (A₁)-(A₃) show the generating bifactor models, (B₁)-(B₃) show the manifest correlation matrices, (C₁)-(C₃) show the ordered alignment ratios (red dashed line marks $\hat{\varphi}^2$; gray dashed line marks the geometric mean of the alignment ratios), and (D₁)-(D₃) show the corresponding two-factor ICM-CFA solutions fitted to the population correlation matrices, with the fitted latent correlations.

that satisfies the proportionality constraint and therefore exhibits alignment-ratio invariance. Panel A₁ shows the generating model, Panel B₁ shows the implied manifest covariance matrix, Panel C₁ shows the ordered alignment ratios (which are all identical), and Panel D₁ shows the corresponding ICM-CFA fit obtained using lavaan (Rosseel, 2012), with the fitted latent correlation reported at 0.6. The second column shows the case of moderate misalignment. The misalignment is most easily seen in Panel C₂, which shows dispersion in the alignment ratios across admissible indices. The geometric-mean reference is shown as a gray dashed line and the fitted $\hat{\varphi}^2$ as a red dashed line, with $\hat{\varphi}^2$ exceeding the geometric mean. The third column shows stronger misalignment, where the gap between $\hat{\varphi}^2$ and the geometric mean increases further. Overall, greater misalignment produces greater upward displacement in the fitted latent correlation.

It is important to emphasize that the demonstrations in Figure 5 are not due to sampling bias. The manifest correlation matrices here are population quantities that hold in the large-sample limit. The results are structural rather than statistical. They show how misalignment displaces the population fitted value from the geometric-mean reference.

A Population-Level Numerical Study

The examples in Figure 5 show the effect of misalignment for three selected population correlation matrices. To assess whether this pattern holds more broadly, we conducted a numerical study at the population level. We generated a large collection of population correlation matrices under the bifactor model with varying degrees of misalignment, fit ICM-CFA directly to each, and compared the fitted latent correlation with the square root of the geometric mean of the alignment ratios.

Each population correlation matrix was generated from a two-cluster orthogonal bifactor model with $J = 10$ tasks, five tasks per cluster, and parameterized through the loading ratios κ_j . Within each cluster,

$$\log \kappa_j = \log \kappa_0 + sz_j,$$

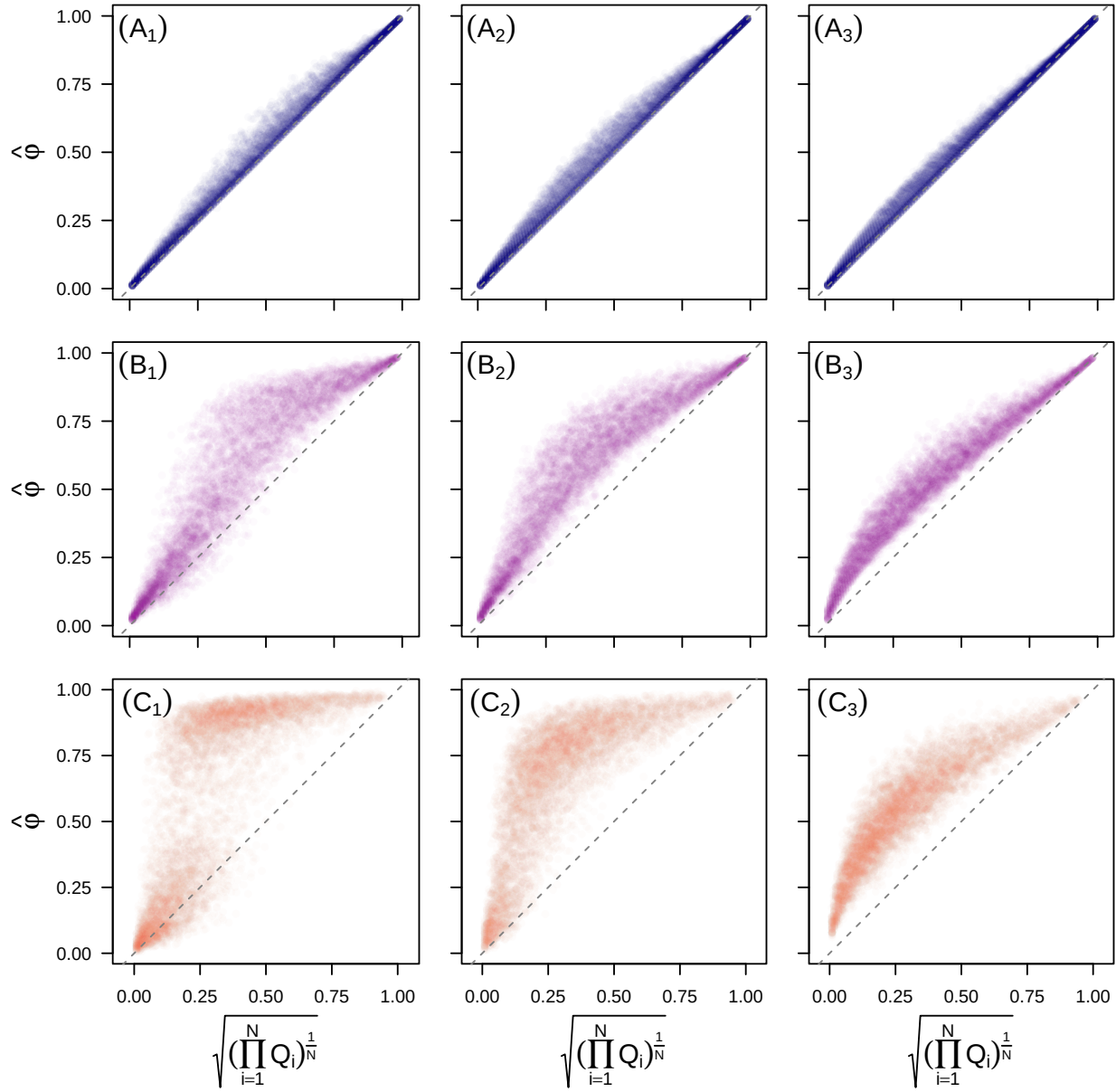
where the z_j were independently drawn from a standard normal distribution and then standardized within the cluster. This construction fixes the geometric mean of the κ_j at κ_0 and their within-cluster log-scale standard deviation at s . Thus, κ_0 fixes the target ratio of specific to general loadings, while s controls the departure from proportionality around that target. To vary the relative contributions of the general and specific factors while holding total common variance constant across tasks, we fixed task communality at $h^2 = 0.5$. Accordingly, $\gamma_j = \sqrt{h^2/(1 + \kappa_j^2)}$ and $\delta_j = \kappa_j \gamma_j$, and unique variance was fixed at 0.5, so $\gamma_j^2 + \delta_j^2 + .5 = 1$ for every task. This construction yields a positive-definite population correlation matrix, so every generated configuration entered the study.³

The numerical design comprised 100 values of κ_0 and 100 values of s . The κ_0 values were selected so that the corresponding proportional-case latent correlations, $\varphi_0 = 1/(1 + \kappa_0^2)$, were equally spaced from .01 to .99. The s values were equally spaced from .01 to 3, spanning near-proportionality to pronounced misalignment.⁴ Ten independent loading configurations were generated for each (κ_0, s) combination, yielding 100,000 population correlation matrices.

For each generated population correlation matrix, we computed the 200 admissible alignment ratios and summarized them by their geometric mean, denoted $\text{GM}(Q)$. We took $\sqrt{\text{GM}(Q)}$ as the population reference on the latent-correlation scale. We then fit ICM-CFA to each population correlation matrix and compared the fitted latent correlation $\hat{\varphi}$ with $\sqrt{\text{GM}(Q)}$. For presentation, configurations were divided into slight, moderate, and high misalignment using terciles of s , corresponding to Rows (A)–(C) of Figure 6. Columns 1 and 2 show normal-theory maximum likelihood (ML) and unweighted least squares (ULS),

³ All code for the numerical study and figures is available at <https://github.com/specl/icm-cfa-latent-cors>.

⁴ We chose the upper limit with the empirical example in mind. Increasing s increases dispersion in the log loading ratios and, in turn, in the population alignment ratios. At $s = 3$, the alignment ratios have an interquartile range of .50 when pooled across κ_0 . This value is substantial but remains below the observed interquartile range of .88 from the Tsukahara et al. example shown in Figure 2(B₃).

**Figure 6**

Fitted latent correlations under misalignment. Scatterplots compare the fitted latent correlation to the square root of the geometric mean of the population alignment ratios, across generated correlation matrices. Rows (A)–(C) correspond to increasing degrees of misalignment (slight, moderate, and high misalignment). Columns 1–3 use the same ICM-CFA specification but with different fitting routines: Column 1 fits the model in lavaan by maximum likelihood, Column 2 fits the same model in lavaan by unweighted least squares, and Column 3 fits the corresponding two-factor model using psych.

respectively, both implemented in `lavaan` (Rosseel, 2012), and Column 3 shows the non-iterative CFA routine in the `psych` package (Revelle, 2026). Across all three fitting routines, the fitted latent correlation moves progressively farther above $\sqrt{\text{GM}(Q)}$ as misalignment increases, although the magnitude of this displacement differs across methods.

Table 1 summarizes the displacement across fitting routines and levels of misalignment. The first numerical column reports relative displacement, defined as $\hat{\varphi}/\sqrt{\text{GM}(Q)}$, where 1 indicates agreement with the population reference. Each entry gives the median across generated population correlation matrices, with the 25th and 75th percentiles in brackets to show the spread across configurations. Median relative displacement is close to 1 under slight misalignment, ranging from 1.03 to 1.04 across fitting routines, but increases to 1.38–1.51 under moderate misalignment and 1.91–2.36 under high misalignment.

The second numerical column reports absolute displacement, defined as $\hat{\varphi} - \sqrt{\text{GM}(Q)}$, where 0 indicates agreement on the latent-correlation scale. Again, the median is reported with the 25th and 75th percentiles in brackets. Median absolute displacement is only 0.01–0.02 under slight misalignment, increases to 0.11–0.15 under moderate misalignment, and reaches 0.24–0.36 under high misalignment. Thus, both summaries show that the fitted latent correlation becomes progressively farther from the population reference as misalignment increases.

Existence and Interpretation of Latent Correlations

The preceding sections show that a correctly specified ICM-CFA model implies alignment-ratio invariance, with every admissible ratio equal to φ^2 . This invariance fails under misassignment, when tasks are assigned to the wrong clusters, or misalignment, when the cluster assignments are correct but the within- and between-cluster covariances fail to align. In either case, the a priori partition does not admit a common alignment ratio. We now consider the implications of this noninvariance.

Table 1

Misalignment	Median ratio	Median difference
ML		
Slight	1.04 [1.01, 1.12]	0.01 [0.00, 0.04]
Moderate	1.38 [1.17, 1.70]	0.12 [0.05, 0.22]
High	1.91 [1.35, 2.80]	0.28 [0.07, 0.50]
ULS		
Slight	1.04 [1.01, 1.15]	0.02 [0.00, 0.04]
Moderate	1.51 [1.25, 1.88]	0.15 [0.08, 0.23]
High	2.36 [1.71, 3.40]	0.36 [0.21, 0.48]
psych		
Slight	1.03 [1.00, 1.11]	0.01 [0.00, 0.03]
Moderate	1.38 [1.16, 1.78]	0.11 [0.07, 0.16]
High	2.09 [1.55, 2.98]	0.24 [0.18, 0.31]

Note. Entries report the median relative displacement and median absolute displacement within each misalignment level, with the 25th and 75th percentiles in brackets.

Existence

By pairwise locality, Equation (8), we can consider the existence of a population latent correlation one cluster pair at a time. For a fixed pair under the *a priori* partition, let $\boldsymbol{\theta} = (\boldsymbol{\Lambda}, \boldsymbol{\Psi}, \varphi)$ collect the parameters of the marginal ICM-CFA model, and let $\boldsymbol{\Sigma}(\boldsymbol{\theta})$ denote its implied covariance matrix, partitioned as in Equation (2). A population latent correlation exists for a given $\boldsymbol{\Sigma}$ only if there is some admissible $\boldsymbol{\theta}$ such that $\boldsymbol{\Sigma}(\boldsymbol{\theta}) = \boldsymbol{\Sigma}$. Under any such exact representation, Equation (7) requires all admissible alignment ratios computed from $\boldsymbol{\Sigma}$ to be invariant with common value φ^2 .

The consequence is immediate. If two admissible index tuples yield different population alignment ratios, they imply different values of φ^2 . No single φ is then compatible with the cluster-pair covariance structure, and without such a φ , no admissible parameter vector can reproduce Σ exactly. Let (ℓ, ℓ', k, k') and $(\tilde{\ell}, \tilde{\ell}', \tilde{k}, \tilde{k}')$ denote any two admissible index tuples for the fixed cluster pair. Then this implication may be written as

$$\begin{aligned} Q(\ell, \ell', k, k') &\neq Q(\tilde{\ell}, \tilde{\ell}', \tilde{k}, \tilde{k}') \\ \implies &\nexists \varphi \text{ compatible with } \Sigma \\ \implies &\nexists \boldsymbol{\theta} \text{ such that } \Sigma(\boldsymbol{\theta}) = \Sigma. \end{aligned}$$

When invariance fails, different admissible combinations of tasks imply different values of φ^2 . A researcher may nonetheless fit an ICM-CFA and obtain $\hat{\varphi}$. But because no single population latent correlation is compatible with the covariance structure, $\hat{\varphi}$ cannot be interpreted as estimating one. Its value instead depends on the criterion used to fit the model to a covariance structure it cannot reproduce.

Fit assessment

Alignment-ratio invariance is a necessary consequence of ICM-CFA. By the existence result above, violations of invariance at the population level preclude an exact representation of Σ under the *a priori* partition and therefore constitute model misspecification. Of course, this misspecification is detectable at some potentially large sample size with a global exact-fit χ^2 test. The misspecification may also be reflected in approximate fit indices, such as CFI and RMSEA, assuming sufficiently large sample sizes.

These assessments are omnibus. The exact-fit test evaluates whether some admissible parameter vector reproduces the full population covariance matrix, while approximate fit indices summarize overall model fit. Information criteria can compare candidate models, but they answer a relative model-selection question rather than identifying a particular failed restriction. Alignment-ratio invariance, by contrast, concerns one latent correlation for one cluster pair. Rejection of the exact-fit null therefore indicates that the model fails somewhere

but does not identify failure of alignment as the source, whereas failure to reject does not establish alignment.

A second concern with omnibus tests is sensitivity. A violation confined to one cluster pair contributes only one component of the overall model discrepancy. As additional tasks and clusters are introduced elsewhere in the model, sensitivity to the same local departure may decrease at a fixed sample size (MacCallum et al., 1996; Satorra & Saris, 1985).

Pairwise locality offers a partial solution by allowing conventional fit assessment to be restricted to one cluster pair at a time. In a K -cluster model, however, doing so entails $\binom{K}{2}$ separate tests and introduces the usual tradeoff between sensitivity and control of false positives across pairs. Even within a pair, the resulting fit test remains omnibus across the restrictions of the two-cluster model. Alignment ratios provide a more targeted diagnostic because they isolate the specific covariance condition required for interpreting the corresponding latent correlation. When substantive inference depends on that correlation, this condition should be evaluated directly.

Consequences

In cognitive research, latent correlations are often treated as central evidence (e.g., Miyake et al., 2000; Friedman & Miyake, 2017), sometimes to the point that manifest correlations receive relatively little interpretive attention.⁵ Their value, especially when large in magnitude, can then take on substantive meaning in its own right. In Tsukahara et al. (2020), for example, the high latent correlation between attentional control and sensory discrimination is used in part by the authors to support the interpretation that attentional control drives performance across sensory-discrimination tasks.

High latent correlations also invite a structural reading. At $\varphi = 1$, the two factors coincide, so a value near one suggests that a single common factor accounts for the covariance

⁵ Manifest correlations are typically reported, but often in tabular form that is difficult to visualize.

among all tasks and that the constructs lack discriminant validity. This reading presupposes that $\hat{\varphi}$ estimates a population latent correlation, which under misalignment it does not. The third column of Figure 5 illustrates this. With $\hat{\varphi} \approx 0.85$, conventional rules of thumb would raise concerns about discriminant validity (Kline, 2011; Shaffer et al., 2016; van Mierlo et al., 2009). Yet the generative model is bifactor, with two cluster-specific factors orthogonal to each other and to the general factor. No single common factor accounts for the covariance among all tasks, the clusters are not redundant, and the high fitted value reflects the misalignment of the ICM-CFA representation rather than redundancy between the clusters.

How are inferences about latent correlations justified? The usual approach has two parts. First, global model fit is assessed using the χ^2 exact-fit test and approximate fit indices such as CFI and RMSEA (Bentler, 1990; Hu & Bentler, 1999; Steiger, 1990). Second, a nested comparison asks whether the two factors need to remain distinct by fixing their correlation to 1 and testing whether fit worsens (Bagozzi et al., 1991). This comparison is conditional on the ICM-CFA representation. It tests whether the common latent correlation can be fixed to 1, not whether the covariance relations support a common latent correlation in the first place. In Tsukahara et al. (2020), a holistic assessment of model fit across multiple indices, together with a nested test fixing the factor correlation to 1, supported a “unity” interpretation of attentional control and sensory discrimination.

The alignment ratios show what these checks leave unresolved. Under ICM-CFA, every population alignment ratio must equal φ^2 and therefore lie in $[0, 1]$. Sampling variability produces dispersion around this common value and may push individual sample ratios beyond either boundary. If population invariance holds, however, the ratios should become increasingly concentrated around φ^2 as sample size grows. Panel (B₃) of Figure 2, based on a sample of 331 participants, shows little such concentration. The ratios span nearly the entire interval from 0 to 1 and extend beyond both boundaries. Although no formal test currently establishes whether this dispersion exceeds that expected from sampling variability

alone, its breadth in a sample of this size provides little descriptive support for a common underlying value and warrants caution when interpreting the fitted latent correlation.

Practical guidance

We have partitioned failure of ICM-CFA into misassignment and misalignment. A conventional approach to misassignment is to use exploratory factor analysis or nonparametric clustering of manifest correlation matrices. If tasks are misassigned, their pattern of correlations with other tasks may not reflect the *a priori* clustering. Researchers can therefore use these approaches as an early diagnostic of possible misassignment. These methods, however, do not address misalignment. A battery may therefore have a plausible cluster structure and still fail the alignment condition that licenses a single latent correlation as a summary of the relations between clusters.

To search for misalignment, we recommend researchers report three descriptive diagnostics alongside the fitted latent correlation $\hat{\varphi}$. First, plot the ordered sample alignment ratios, as in Figure 2; departure from a flat pattern displays disagreement across admissible combinations of tasks. Second, report their interquartile range as a compact measure of dispersion. Third, when all ratios are positive, compare $\hat{\varphi}$ with the square root of their geometric mean. In the numerical study, the fitted value tended to exceed this reference under nontrivial misalignment. Their difference summarizes the separation between the fitted parameter and the geometric-mean reference.

These diagnostics are descriptive rather than inferential. Empirical variation in the ratios reflects both population misalignment and sampling variability, and no distributional calibration is currently available. They complement rather than replace exploratory analyses and conventional model-fit assessment. A formal inferential treatment that calibrates sampling variability and accounts for the dependence among ratios is beyond the scope of the present article and remains a topic for future work.

Discussion

This paper identifies the covariance restriction required for a single latent correlation to summarize the relations between two clusters. The alignment ratios computed from the manifest covariances must be invariant with a common value of φ^2 . Misassignment and misalignment both break this invariance, and when the invariance fails then no single population latent correlation exists for the cluster pair. Because the alignment ratios are computed directly from the manifest covariance matrix, they provide a descriptive diagnostic that may be reported alongside any fitted latent correlation without additional modeling.

Alternatives to ICM-CFA

We have focused on ICM-CFA because it is perhaps the most popular model for assessing latent correlations. Yet, there are several alternatives to ICM-CFA. One is to retain a cluster structure but relax the constraint that tasks load on one and only one factor. The resulting model is often called the exploratory structural equation model (ESEM, Asparouhov & Muthén, 2009; Marsh et al., 2014; Morin et al., 2016). Allowing cross-cluster loadings reduces the need for the model to absorb cross-cluster covariance into the latent correlation. As a result, ESEM models attenuate the inflation mechanisms described here, a fact that is emphasized in the ESEM literature. ESEM trades this benefit, however, for additional flexibility and ambiguity. Factor definitions depend on rotation and target specifications, and the resulting factor correlations may be sensitive to these choices. This flexibility can also weaken the simple cluster-based interpretation that motivates ICM-CFA in cognitive batteries, where factors are intended to map cleanly onto named constructs. For this reason, ESEM has been less widely adopted in cognitive-battery applications, despite its appeal as a remedy for cluster misassignment. Moreover, permitting cross-loadings directly addresses misassignment-type misspecification, but it does not guarantee alignment of within-cluster and between-cluster submatrices.

Another alternative is the bifactor model, in which tasks load on both a general

factor and a cluster-specific factor (Holzinger & Swineford, 1937; Reise, 2012). Bifactor models can accommodate misalignment because the relative sizes of these loadings need not be the same across tasks. When proportionality holds, the relation between two clusters can be reduced to a single latent correlation through Equation (9). With more than two clusters, extending this reduction requires rank-one structure among the latent correlations and leads to a hierarchical CFA representation (Brunner et al., 2012; Schmid & Leiman, 1957; Yung et al., 1999). Proportionality, however, is not required for a bifactor model to be interpretable. Its general and cluster-specific factors can be meaningful without it (Eid et al., 2017; Reise, 2012), and omega coefficients summarize how observed-score variance is distributed among them (Rodriguez et al., 2016). These interpretations do not require the covariance relations between clusters to reduce to a single latent correlation. They do, however, depend on whether the factors are appropriately defined by the underlying random experiment; when these requirements are not met, bifactor models may yield anomalous results and do not constitute a valid alternative to ICM-CFA (Eid et al., 2017).

Future Directions

Two future directions follow. First, it would be useful to develop formal methods for testing alignment-ratio invariance under sampling variability. One approach is to build test statistics in the spirit of vanishing tetrads and related constraint tests in factor analysis and SEM (Bollen & Ting, 1993; Drton et al., 2007; Wishart, 1928), complemented perhaps by bootstrap procedures that assess the stability of invariance across resamples. Second, the alignment-ratio framework suggests a constructive route to cluster discovery. One could search for cluster partitions that yield approximately invariant alignment ratios, so that the resulting oblique factor structure admits a sound latent correlation. This could be implemented with combinatorial optimization. For example, one could minimize a dispersion measure of the alignment ratios (or a direct invariance criterion) over candidate partitions.

Funding: MM and JNR were supported by ONR N00014-23-1-2792.

Conflict of Interest: The authors declare no competing interests.

Ethical Approval: NA

Consent to Participate: NA

Consent for Publication: NA

Data Availability: NA

Use of AI tools: We used an LLM for copyediting (grammar and clarity).

Code Availability: All code for the analyses and figures is available at <https://github.com/specl/icm-cfa-latent-cors>.

Author Contributions: MM and JNR conceived the study and wrote the manuscript. MM derived the theoretical results, implemented the simulations, produced the figures, and conducted the analyses. JNR provided supervision and critical revisions throughout.

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